



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

AM2001-1 – ADVANCED MACHINE LEARNING
 (AI&ML)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is one-hot encoding used for?**
 - A) Handling missing values
 - B) Encoding categorical variables
 - C) Scaling numerical features
 - D) Feature selection
2. **Which method is commonly used for imputing missing numerical values?**
 - A) Mean imputation
 - B) Mode imputation
 - C) Median imputation
 - D) Random imputation
3. **Why is feature scaling important in machine learning?**
 - A) To handle missing values
 - B) To speed up the training process
 - C) To standardize the range of features
 - D) To improve model interpretability
4. **Logistic Regression is used for**
 - A) Regression problems
 - B) Binary classification problems
 - C) Clustering
 - D) Dimensionality reduction
5. **What is hyperparameter tuning?**
 - A) Selecting the best features
 - B) Fine-tuning the model parameters
 - C) Handling missing values
 - D) Encoding categorical variables
6. **What is BeautifulSoup used for in web scraping?**
 - A) Extracting features from data
 - B) Parsing and navigating HTML content
 - C) Handling missing values in web data
 - D) Training machine learning models
7. **In BeautifulSoup, which method is used to find all instances of a particular HTML tag?**
 - A) find_all
 - B) find
 - C) locate_all
 - D) locate
8. **What is the purpose of web crawling?**
 - A) Extracting information from websites
 - B) Analyzing social media data
 - C) Cleaning and preprocessing data
 - D) Deploying machine learning models
9. **Which Python library is commonly used for building web crawlers and spiders?**
 - A) BeautifulSoup
 - B) Pandas
 - C) Scrapy
 - D) Requests
10. **What is the role of a web spider in web crawling?**
 - A) It captures images from websites
 - B) It removes irrelevant content from web pages
 - C) It encrypts web communication
 - D) It navigates through web pages and collects data
11. **What is computer vision?**
 - A) Teaching computers to see and interpret visual information
 - B) Enhancing computer processing speed
 - C) Developing virtual reality applications
 - D) Implementing encryption algorithms
12. **What does the Harris Corner Detection algorithm identify in images**
 - A) Edges
 - B) Textures
 - C) Corners
 - D) Colors

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13. In Min-Max scaling, what is the range of the scaled values
 A) 0 to 1 B) -1 to +1
 C) 1 to 100 D) None of the above
14. What is the sigmoid function used for in logistic regression?
 A) Feature scaling B) Predicting probabilities
 C) Handling missing values D) One-hot encoding
15. In a machine learning pipeline, what is the purpose of the 'fit_transform' method?
 A) Fitting the model B) Transforming the data
 C) Both a and b D) Neither a nor b
16. What is grid search used for in hyperparameter tuning?
 A) Randomly selecting hyperparameters B) Evaluating all possible combinations of hyperparameters
 C) Ignoring hyperparameters D) Modifying the training dataset
17. What is the kernel trick used for in Support Vector Machines?
 A) Encoding categorical variables B) Dimensionality reduction
 C) Non-linear mapping in feature space D) Handling missing values
18. Which of the following is true about robots.txt in the context of web scraping?
 A) It provides information about website owners B) It is used for encryption
 C) It specifies rules for web crawlers D) It contains HTML code snippets
19. Which of the following is a characteristic of a well-behaved web spider?
 A) Ignoring 'robots.txt' rules B) Crawling too quickly and frequently
 C) Extracting irrelevant information D) Being respectful of website resources and policies
20. What does SIFT stand for in computer vision?
 A) Scale-Invariant Feature Technique B) Significant Image Feature Transform
 C) Stable Image Feature Tracker D) Scale-Invariant Feature Transform

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Elaborate on categorical encoding with different methods with sample code.	8	3	1	2
	b) How do different scaling methods help in data preprocessing? Explain any two methods with sample code snippet.	8	3	1	2
2.	a) Elaborate the concept of data pipelines and its usefulness with example.	8	3	1	2
	b) Write a short note on Ensemble hyperparameter tuning techniques.	8	2	2	1
3.	a) Illustrate the training process of logistic regression.	8	2	2	1
	b) Elaborate on different kernels of support vector machines and state its usefulness with code snippet.	8	3	2	2
Unit – II					
4.	a) Explain different web scrapping techniques used for scrapping data.	8	2	3	1
	b) Illustrate with an example, find() and find_all() methods with BeautifulSoup().	8	2	3	1
5.	a) Explain commonly used Regular Expressions with sample code.	8	3	3	2
	b) Write web crawling process and a simple program to illustrate this.	8	3	4	2
6.	a) Write the key components of scrapy.	8	2	4	1
	b) Define scrapy and illustrate how to write simple scraper.	8	3	4	2
Unit – III					
7.	a) Illustrate interactive annotations with a sample code.	8	3	5	2
	b) Write the steps involved in principle component analysis apply on images.	8	2	5	1
8.	a) Illustrate Harris -Corner detector with an example.	8	3	5	1
	b) Explain matching geotagged images.	8	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
